Data Quality Report

Before = raw source files. After = loaded warehouse.

Source	Metric	Before	After
Book loans	Rows	3025	3000
Book loans	Duplicate rows	25	0
Book loans	Department spellings	20	7
Book loans	Empty department	106	0
Book loans	Dates not in YYYY-MM-DD	872	0
Book loans	Empty ReturnDate (kept, not returned yet)	487	483
Digital downloads	Rows	2100	2100
Digital downloads	Resource type spellings	6	3
Digital downloads	Faculty spellings	20	7
Digital downloads	DownloadCount with commas	20	0
Digital downloads	Missing Duration_Minutes (kept as NULL)	500	500
Room bookings	Rows	1540	1500
Room bookings	Duplicate rows	40	0
Room bookings	Room number spellings	30	10
Room bookings	Time slot spellings	11	3
Room bookings	Walk-ins without StudentID (kept as Unknown)	126	121
Warehouse	Broken foreign keys	n/a	0

This report compares the raw source files with the loaded warehouse. The data used is sample data generated for this project, not real library records. The three biggest improvements are: department names were reduced from 20 spellings to 7, 872 dates in other formats were converted to YYYY-MM-DD, and 65 duplicate records were removed. Some numbers are not zero on purpose: 483 books are still not returned and 121 walk-in room bookings are kept under an Unknown student, because deleting them would understate library usage. One limitation is that 500 download records have no reading time, which cannot be recovered, so they are stored as empty.